Name: Class:

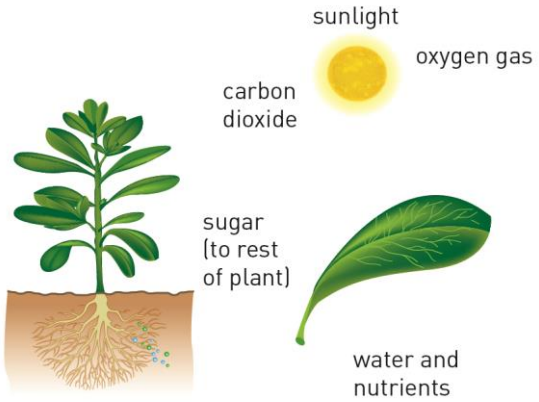
Student worksheet

6.2 All organisms have a role in an ecosystem

Pages 104–105 and 199

Ecosystems – energy for life

1 In most ecosystems, the Sun provides the energy for the producers. The producers convert the Sun's energy into food through a process called photosynthesis. During this process, sunlight shining on the green leaf is converted to chemical energy by chlorophyll contained in the leaves, and is then used to convert carbon dioxide and water into sugar and oxygen. Use this description to label the arrows on the diagram.



2 You may remember this description from the previous worksheet:

The following observations were made while sitting by a billabong.

Nymphs, water beetles and fish were observed eating algae and water plants. Dragonflies were seen consuming the water beetles and nymphs. A large black swan started feeding on the water plants, but also ate the nymphs and water beetles. A long-necked turtle also ate the water plants and water beetles. Then, a huge wedged-tailed eagle swooped down and plucked the turtle from the water and started to eat it.

a Which consumers are herbivores?

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b Which consumers are carnivores?

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c Which consumers are omnivores?

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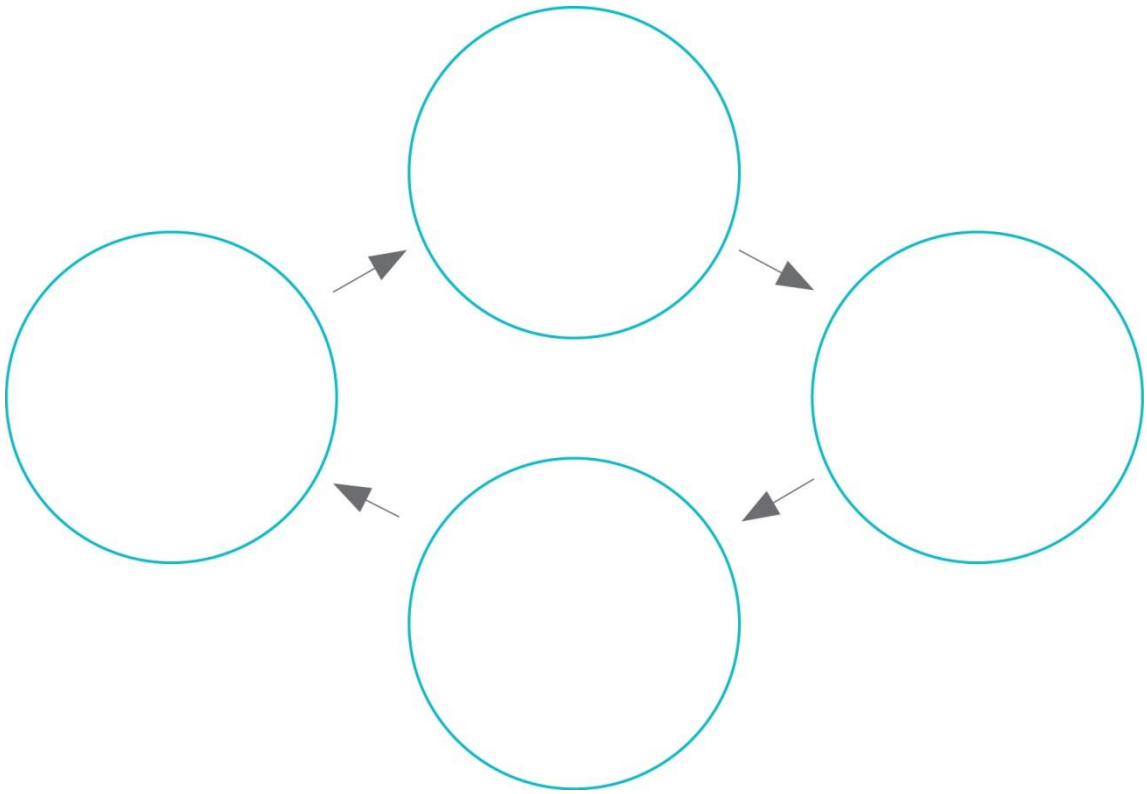
Name: Class:



3 The cow in the image is in the process of being recycled by decomposers.

Fill in the flow chart below to show the cycle of nutrients and energy that this dead cow is part of in the ecosystem. Include the following information:

- the producer, consumers and decomposer
- carnivore and herbivore
- an example of the producer, consumer (other than the cow) and the decomposer.



EXTEND YOUR UNDERSTANDING

4 Find out how a worm decomposes dead organisms and how this is linked to the health of the soil in a garden. The following website may be a useful place to start:
http://www.dpi.nsw.gov.au/agriculture/resources/soils/biology/earthworms

You could present your findings as a one- to two-paragraph written response. A diagram may also be worthwhile.

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